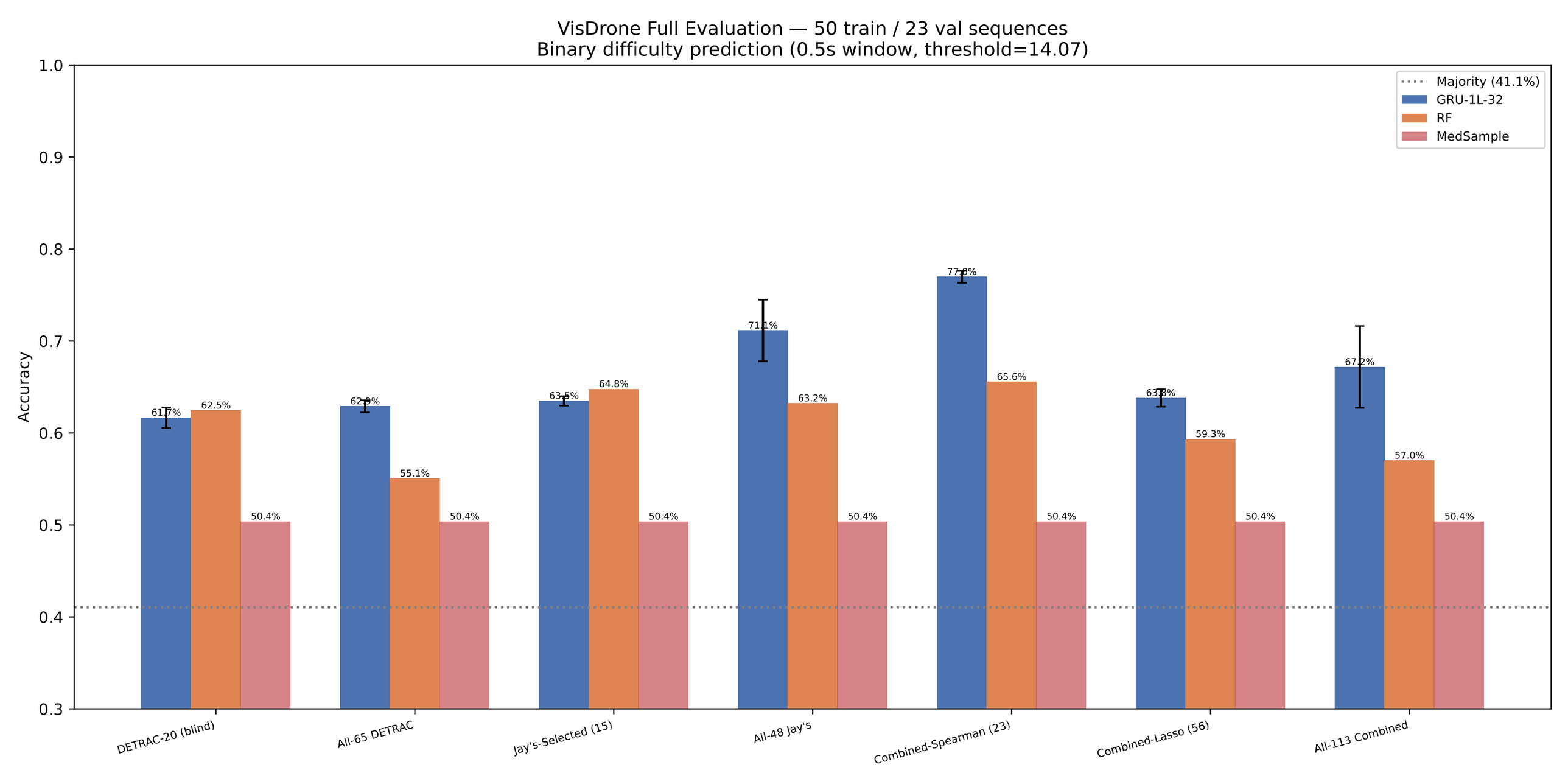
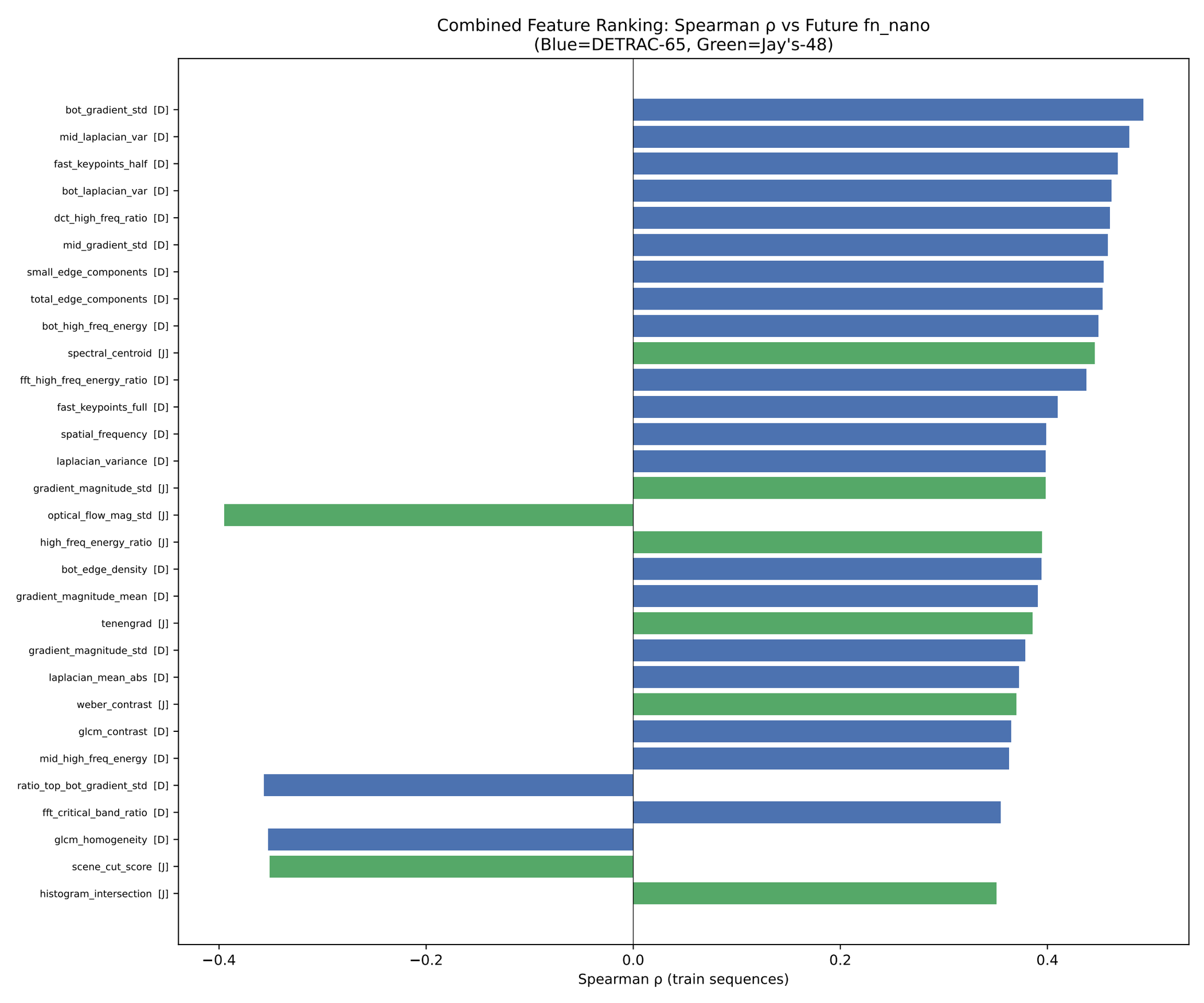
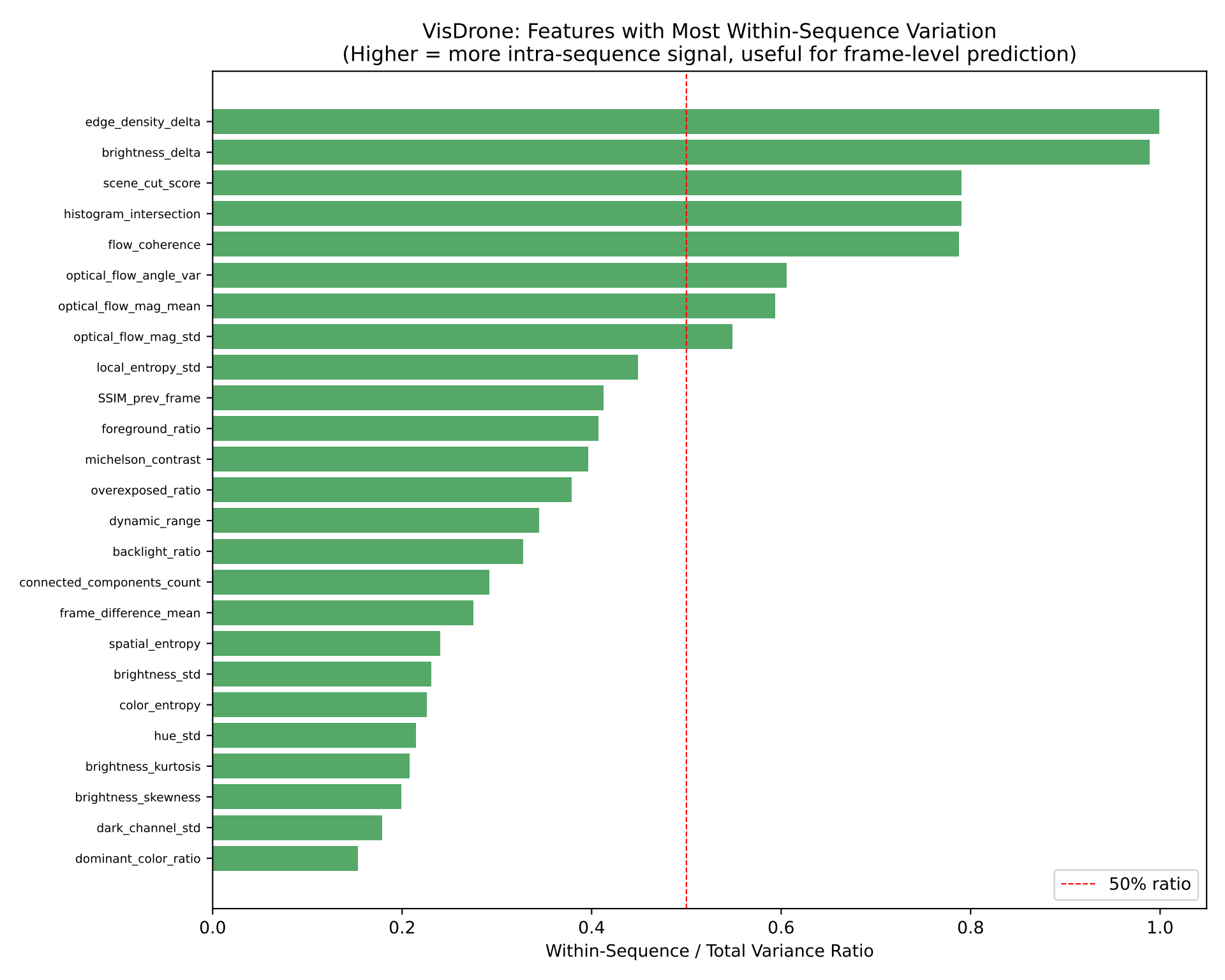
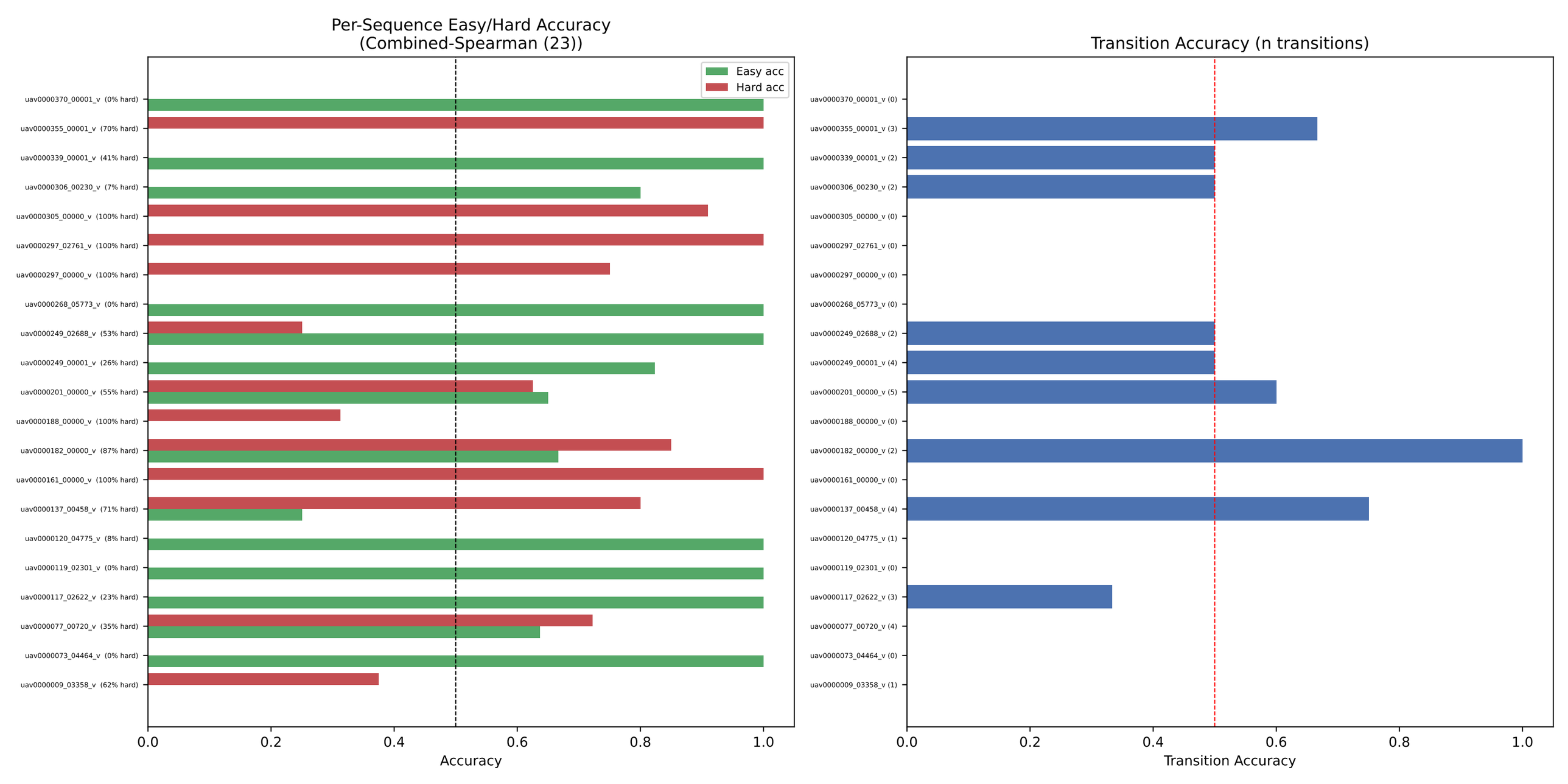

VisDrone Full Evaluation — 50 train / 23 val sequences
Binary difficulty prediction (0.5s window, threshold=14.07)
Accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
Majority (41.1%)
GRU-1L-32
RF
MedSample
62.5%
50.4%
55.1%
50.4%
63.5%
64.8%
50.4%
71.1%
63.2%
50.4%
65.6%
50.4%
59.3%
50.4%
67.2%
57.0%
50.4%
DETRAC-20 (blind)
All-65 DETRAC
Jay's-Selected (15)
All-48 Jay's
Combined-Spearman (23)
Combined-Lasso (56)
All-113 Combined


Combined Feature Ranking: Spearman ρ vs Future fn_nano
(Blue=DETRAC-65, Green=Jay's-48)
bot_gradient_std [D]
mid_laplacian_var [D]
fast_keypoints_half [D]
bot_laplacian_var [D]
dct_high_freq_ratio [D]
mid_gradient_std [D]
small_edge_components [D]
total_edge_components [D]
bot_high_freq_energy [D]
spectral_centroid [J]
fft_high_freq_energy_ratio [D]
fast_keypoints_full [D]
spatial_frequency [D]
laplacian_variance [D]
gradient_magnitude_std [J]
optical_flow_mag_std [J]
high_freq_energy_ratio [J]
bot_edge_density [D]
gradient_magnitude_mean [D]
tenengrad [J]
gradient_magnitude_std [D]
laplacian_mean_abs [D]
weber_contrast [J]
glcm_contrast [D]
mid_high_freq_energy [D]
ratio_top_bot_gradient_std [D]
fft_critical_band_ratio [D]
glcm_homogeneity [D]
scene_cut_score [J]
histogram_intersection [J]
−0.4
−0.2
0.0
0.2
0.4
Spearman ρ (train sequences)


VisDrone: Features with Most Within-Sequence Variation
(Higher = more intra-sequence signal, useful for frame-level prediction)
edge_density_delta
brightness_delta
scene_cut_score
histogram_intersection
flow_coherence
optical_flow_angle_var
optical_flow_mag_mean
optical_flow_mag_std
local_entropy_std
SSIM_prev_frame
foreground_ratio
michelson_contrast
overexposed_ratio
dynamic_range
backlight_ratio
connected_components_count
frame_difference_mean
spatial_entropy
brightness_std
color_entropy
hue_std
brightness_kurtosis
brightness_skewness
dark_channel_std
dominant_color_ratio
50% ratio
0.0
0.2
0.4
0.6
0.8
1.0
Within-Sequence / Total Variance Ratio


Per-Sequence Easy/Hard Accuracy
(Combined-Spearman (23))
Easy acc
Hard acc
uav0000370_00001_v (0% hard)
uav0000355_00001_v (70% hard)
uav0000339_00001_v (41% hard)
uav0000306_00230_v (7% hard)
uav0000305_00000_v (100% hard)
uav0000297_02761_v (100% hard)
uav0000297_00000_v (100% hard)
uav0000268_05773_v (0% hard)
uav0000249_02688_v (53% hard)
uav0000249_00001_v (26% hard)
uav0000201_00000_v (55% hard)
uav0000188_00000_v (100% hard)
uav0000182_00000_v (87% hard)
uav0000161_00000_v (100% hard)
uav0000137_00458_v (71% hard)
uav0000120_04775_v (8% hard)
uav0000119_02301_v (0% hard)
uav0000117_02622_v (23% hard)
uav0000077_00720_v (35% hard)
uav0000073_04464_v (0% hard)
uav0000009_03358_v (62% hard)
0.0
0.2
0.4
0.6
0.8
1.0
Accuracy
Transition Accuracy (n transitions)
uav0000370_00001_v (0)
uav0000355_00001_v (3)
uav0000339_00001_v (2)
uav0000306_00230_v (2)
uav0000305_00000_v (0)
uav0000297_02761_v (0)
uav0000297_00000_v (0)
uav0000268_05773_v (0)
uav0000249_02688_v (2)
uav0000249_00001_v (4)
uav0000201_00000_v (5)
uav0000188_00000_v (0)
uav0000182_00000_v (2)
uav0000161_00000_v (0)
uav0000137_00458_v (4)
uav0000120_04775_v (1)
uav0000119_02301_v (0)
uav0000117_02622_v (3)
uav0000077_00720_v (4)
uav0000073_04464_v (0)
uav0000009_03358_v (1)
0.0
0.2
0.4
0.6
0.8
1.0
Transition Accuracy

Full Evaluation Summary — VisDrone (50 train / 23 val)

Feature Set	GRU (3-seed)	RF	MedSample	Maj.	Intra Signal	Trans Acc	n_val
DETRAC-20 (blind)	61.7% ± 1.1%	62.5%	50.4%	41.1%	1/12	45.5% (33t)	514
All-65 DETRAC	62.9% ± 0.7%	55.1%	50.4%	41.1%	3/12	54.5% (33t)	514
Jay's-Selected (15)	63.5% ± 0.5%	64.8%	50.4%	41.1%	3/12	57.6% (33t)	514
All-48 Jay's	71.1% ± 3.3%	63.2%	50.4%	41.1%	3/12	54.5% (33t)	514
Combined-Spearman (23)	77.0% ± 0.6%	65.6%	50.4%	41.1%	3/12	48.5% (33t)	514
Combined-Lasso (56)	63.8% ± 1.0%	59.3%	50.4%	41.1%	4/12	54.5% (33t)	514
All-113 Combined	67.2% ± 4.4%	57.0%	50.4%	41.1%	4/12	51.5% (33t)	514